

COSC 1336 – Fundamentals of Programming

Project Assignment #9 - Car Mileage

Objective:

Write a program that defines and uses a **Car class** to simulate a car's behavior in terms of acceleration and braking.

Class Definition: Car

The class should include the following **member variables**:

- year: an int representing the car's model year.
- make: a string representing the make of the car.
- speed: an int representing the car's current speed (initialized to 0).

Member Functions:

1. **Constructor:**
 - Accepts the car's **year** and **make** as arguments.
 - Initializes the year and make with the given values.
 - Sets the initial speed to **0**.
2. **Accessors:**
 - Provide accessor methods (getYear(), getMake(), getSpeed()) to retrieve the values of the member variables.
3. **accelerate() Method:**
 - Adds **5** to the speed each time it is called.
4. **brake() Method:**
 - Subtracts **5** from the speed each time it is called.

Program Requirements:

- Prompt the user to enter the **year** and **make** of the car.
- Ask the user how many times the car should **accelerate** and **brake**.
- Create an instance of the Car class using the input values.
- Call the accelerate() method the specified number of times, displaying the **current speed** after each call.
- Call the brake() method the specified number of times, again displaying the **current speed** after each call.

Required Functions:

The program **must use** the following functions:

- getStringData()
 - Validates and returns a non-empty string for user input (e.g., make of the car).
- getIntegerData()
 - Validates and returns a **positive integer** (e.g., number of times to accelerate/brake).
- displayResult()
 - Displays the current speed after each acceleration or brake operation.
- main()
 - Manages the flow of the program and coordinates all function calls.

Additional Guidelines:

- Validate all user inputs.

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- Use **professional programming style**: proper naming conventions, indentation, and documentation.
- Include descriptive **comments** at the beginning of the file and before each function.

Sample Output (User entry is in **RED**)

Project #9
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My Car Information

Car Make: **Chevy**
Enter year: **2022**

How many times did you accelerate? **6**
How many times did you brake? **2**

Car Travel Information

Acceleration applied:

Accelerate #1:	Increase by 5 mph	Current speed: 5 mph
Accelerate #2:	Increase by 5 mph	Current speed: 10 mph
Accelerate #3:	Increase by 5 mph	Current speed: 15 mph
Accelerate #4:	Increase by 5 mph	Current speed: 20 mph
Accelerate #5:	Increase by 5 mph	Current speed: 25 mph
Accelerate #6:	Increase by 5 mph	Current speed: 30 mph

Brake Applied

Brake #1:	decrease by 5 mph	Current speed: 25 mph
Brake #2:	decrease by 5 mph	Current speed: 20 mph

Chevy Final Speed: 20 mph

End of Project 9